

- 1 A student investigated the effect of gibberellin (GA) and abscisic acid (ABA) on the germination of wheat grains.

The student found out the following information.

- The molecular mass of GA is 346 g.
- The molecular mass of ABA is 264 g.
- The range of concentrations at which GA is active in plant tissues is from 1×10^{-6} to $1 \times 10^{-5} \text{ mol dm}^{-3}$.
- The range of concentrations at which ABA is active in plant tissues is from 1×10^{-6} to $1 \times 10^{-4} \text{ mol dm}^{-3}$.
- Newly formed wheat grains contain a high concentration of ABA which decreases over time.
- When the concentration of ABA falls low enough the GA is able to promote germination.
- Wheat germinates best at temperatures from 10°C to 25°C and without light.

Fig. 1.1 shows the germination and early growth of wheat.

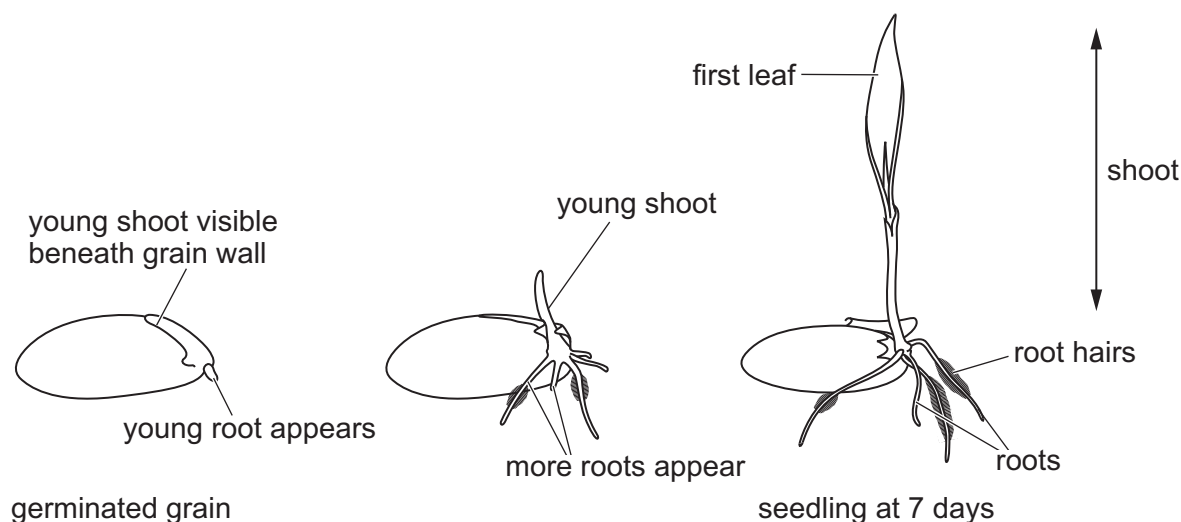


Fig. 1.1

To test the effects on germination of adding ABA, the student started by soaking wheat grains for 24 hours in a solution of $1 \times 10^{-6} \text{ mol dm}^{-3}$ GA. These were then split into several groups of 50 grains and a different concentration of ABA was added to each group.

The student made a stock solution of ABA with a concentration of $1 \times 10^{-3} \text{ mol dm}^{-3}$ which was then diluted to make the solutions needed for the investigation.

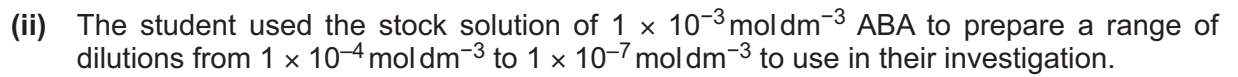
- (a) (i) Using the information about ABA above, describe how the student made a stock solution of $1 \times 10^{-3} \text{ mol dm}^{-3}$ ABA.

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Describe how the student prepared these dilutions.

(b) (i) Identify the independent and the dependent variables in this investigation.

dependent variable..... [2]

- Your method should be detailed enough for another person to use and should **not** repeat any details from **(a)(ii)** of how to dilute the stock solution.

[5]

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100

(c) The student calculated the percentage germination of the grains in the different concentrations of ABA.

(i) Explain how the percentage germination was calculated.

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..... [1]

(ii) Complete Fig. 1.2 by sketching a graph to predict the results the student might expect. On the axes, show the range of values and the units that might be used to plot the graph.



Fig. 1.2

[3]

- (d) In another investigation the student tested the effect of gibberellin (GA) on the elongation of shoots. Five groups of 20 wheat grains were soaked in GA solution. Each group was treated with a different concentration of GA and then left to germinate and grow for 10 days. The length of each shoot was measured and a mean shoot length was calculated for each GA concentration.

Table 1.1 shows the results of this investigation.

Table 1.1

concentration of GA/mol dm ⁻³				
0	1×10^{-7}	1×10^{-6}	1×10^{-5}	1×10^{-4}
mean shoot length/mm $\pm$ SE				
120 ± 10	150 ± 5	175 ± 15	210 ± 12	180 ± 8

- (i) Explain what standard error (SE) shows about the reliability of the results in Table 1.1.

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 [2]

The student carried out a number of *t*-tests.

In each *t*-test, the student compared the mean shoot length of the seedlings grown from grains that were **not** treated with GA with those that were treated with GA.

- (ii) Suggest a null hypothesis for the *t*-test between **no** GA treatment and those treated with 1×10^{-5} mol dm⁻³ GA.

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 [1]

- (iii) Explain how the student should use the values for *t* to determine if there is a significant difference between the mean shoot length of wheat grown with **no** GA treatment and the mean shoot length of wheat grown with GA treatment.

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 [3]

[Total: 21]